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A novel data-driven graph-based requirement elicitation framework in the smart product-service system context

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ABSTRACT

Nowadays, industrial companies are facing ever-increasing challenges to generate new value-in-use and maintain their high competitiveness in the market. With the rapid development of Information and Communication Technology (ICT), IT is embedded in the products themselves, i.e. smart, connected products (SCPs) to generate values. Hence, an emerging value proposition paradigm, smart product-service system (Smart PSS) was introduced, by leveraging both SCPs and its generated services as a solution bundle to meet individual customer needs. Unlike other types of PSS, in Smart PSS, massive user-generated data and product-sensed data are collected during the usage phase, where potential requirements can be elicited readily in a value co-creation manner with context-awareness. Nevertheless, only few scholars discuss any systematic manner to support requirement elicitation in such context. To fill the gaps, this research proposes a novel data-driven graph-based requirement elicitation framework in the Smart PSS, so as to assist engineering/designers make better design improvement or new design concept generation in a closed-loop manner. It underlines the informatics-based approach by integrating heterogeneous data sources into a holistic consideration. Moreover, by leveraging graph-based approach, context-product-service information can be linked by the edges and nodes in-between them to derive reliable requirements. To validate its feasibility and advantages, an illustrative example of smart bicycle design improvement is further adopted. As an explorative study, it is hoped that this work provides useful insights for the requirement elicitation process in today's smart connected environment.

1. Introduction

The rapid changing global market requires the transition of industrial business model from solely product sales to product-service system (PSS), as an integrated bundle including both physical products and intangible services, to meet individual customer needs [1,2]. With the third wave of IT innovation, cutting-edge technologies (e.g. Internet of Things (IoT), cyber-physical systems (CPS)) are revolutionizing the PSS into a novel paradigm, i.e. Smart PSS [2]. It is endowed with unique features including connectivity, integration, autonomy and digitalization [3], and has shown its uniqueness in its solution design process [4]. Owing to its great advantages, numerous services can be generated throughout the entire engineering lifecycle, such as conceptual design based on user behaviors [5], manufacturing planning based on real-time data [5], remote machine health monitoring [6] and business innovations [7], leading to a more demanding and complex system which aims to fulfill diverse requirements.

As a result, the success of Smart PSS relies much on the quality of product-service bundles that to what extent the system satisfies the users' requirements [8]. However, the conventional requirements elicitation techniques, like interview, observation, questionnaire or focus group, are mostly time-consuming and human-intensive in nature, and hence not suitable for the Smart PSS domain [9]. Furthermore, the vagueness and even false information happen when using the conventional requirement elicitation techniques due to the subjective assessment of people. Facing this problem, adding multiple sources of reliable data (e.g. contextual data collected by sensors) can alleviate the phenomenon. In Smart PSS, both user-generated data and product-sensed data can be accessible by leveraging the large-scale SCPs in a networked environment [10]. Moreover, with the support of Internet and the global market, the users are usually geographically distributed with different backgrounds and profiles [11], causing it hard to identify stakeholders and their requirements. Hence, there is a need to develop a systematic framework for requirement elicitation in today's smart

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connected environment [12,13]. Aiming to fill aforementioned gaps, this research establishes a generic framework to guide the process of requirement elicitation by leveraging massive user-generated and product-sensed data from SCPs in the context of Smart PSS.

The rest of this paper is organized as follows. Section 2 gives a holistic review of related works. Section 3 represents the proposed data-driven and graph-based requirement elicitation framework in the Smart PSS. In Section 4, an illustrative example of smart bicycle design improvement is given to validate the feasibility and advantages of the proposed framework. Advantages and limitations of the framework and example are further discussed in Section 5. At last, Section 6 concludes the main contributions of this research and highlights the future work.

2. Literature review

This section aims to clarify the distinctions between current PSS paradigms and the emerging Smart PSS, as well as the issues in its requirement elicitation tasks. Related works on the PSS evolvement from both technical aspect and organizational aspect, and existing requirement elicitation methods are comprehensively reviewed below.

2.1. Evolvement of IT-driven PSS

PSS was first coined by Goedkoop et al. [14] as “a system of products, services, networks of players and supporting infrastructure that continuously strives to be competitive, satisfy customer needs and have a lower environmental impact than traditional business models.” Except for the novel business model, today’s companies are largely shifting their value proposition from a product-centric one to a product-service integrated one, by embracing the technical innovations as well. According to the study of Zheng et al. [4], the IT-driven PSS paradigms can be generally classified into three categories, namely conventional PSS, IoT-enabled PSS and Smart PSS, from the perspective of techniques applied (shown in Fig. 1). The main distinction between conventional PSS and IoT-enabled PSS is driven by the pervasive usage of IoT technology which enables the ubiquitous connection and online intelligence. The key technical progress of Smart PSS is the application of smart connected products (SCPs), which are embedded with microchips, software and sensors so that they are capable of collecting, processing and produce information and even have the ability of autonomy. Meanwhile, the achievement of cyber-physical system enables the seamless integration and interaction between both physical products and its generated e-services in the cyber space. Based on this taxonomy, technical PSS [15] and industrial PSS [16] are classified as the conventional PSS. Meanwhile, IoT-enabled PSS are the ones enabled by the ubiquitous connectivity of IoT with online intelligence, and cyber-physical PSS [17], digitalized PSS [18] and Smart PSS [19] are grouped as Smart PSS by considering both cyber and physical spaces.

Except for the technology transformation of Smart PSS, the organizations and activities in Smart PSS are evolved as well. A more intelligent design manner, i.e. design-wise [20] is expected. First of all, Smart PSS integrates the SCPs and its generated e-services holistically by combining multi-disciplinary engineering fields together [21], making the system as a complex system in the smart, connected environment. Secondly, large number of e-services and advanced IT enable designers to implement value co-creation interactions with end-users along the solution design process [20]. Hence, more intelligent decisions can be made with better user experience and satisfaction. Furthermore, the design of Smart PSS value proposition is a high context-dependent activity, in which the context (e.g. the market, end goal, scenarios) should be carefully considered for value generation [22].

2.2. Related works and tools in requirement elicitation

The task of requirement elicitation can be regarded as the process of identification of stakeholders and extracting requirements from stakeholders and other relevant resources [9]. Fernandes and Machado [23] have addressed that the main steps of requirement elicitation contain *study domain, identify sources, consult stakeholders, select techniques and elicit requirements*.

In order to understand the domains, ontology is a widely-accepted way which can formally represent knowledge as a set of concepts in a domain, using a shared vocabulary to denote types, relationship, properties and so on [24]. It has the advantages of readability and processability [25] that ontologies can store knowledge with additional annotations using RDF, a machine-understandable format, and it can support automatic reasoning and traceability of requirements, which improve the reusability of requirements. Castañeda et al. [26] have summarized that the ontologies have three kinds of roles in requirement engineering, i.e. serving as the representation model of requirements, as the knowledge structure of a specific domain and to represent the domain knowledge. Correspondingly, they defined three types ontologies in requirement engineering, namely requirement ontology, requirement specification document ontology and application domain ontology. Furthermore, to achieve the mapping between requirements and ontological concepts, Kitamura et al. [27] developed a requirement elicitation tool called ORE (Ontology driven Requirements Elicitation), which can extract items (in textual format) from requirement documents and match them with ontological concepts.

As for the requirement elicitation reasoning methods themselves, diverse types of methods are applied, including multi-criteria decision-making methods and machine-learning algorithms. For example, Sousa-Zomer and Miguel [28] combined quality function deployment (QFD) and fuzzy analytic hierarchy process (FAHP) to analyze and prioritize stakeholders’ requirements. Koitz and Glinz [11] built fuzzy Galois lattices to analyze user requests and to generate optimum solutions for

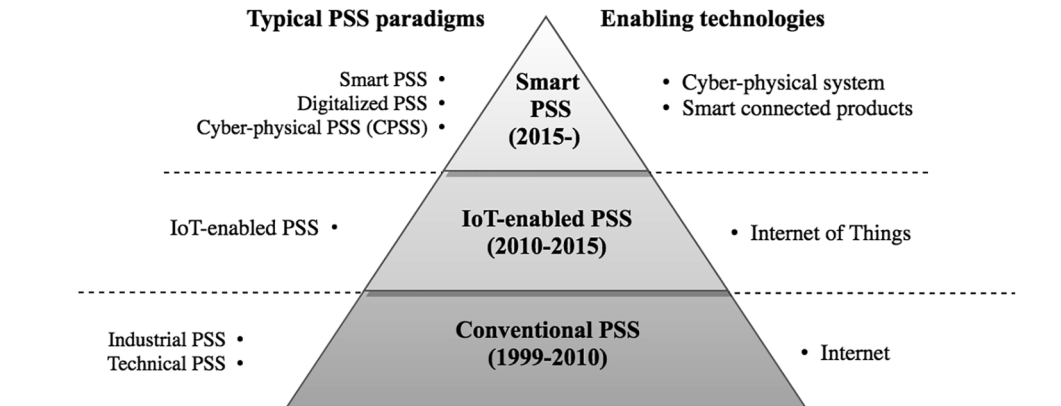


Fig. 1. The evolvement of PSS from the technical aspect (derived from [4]).

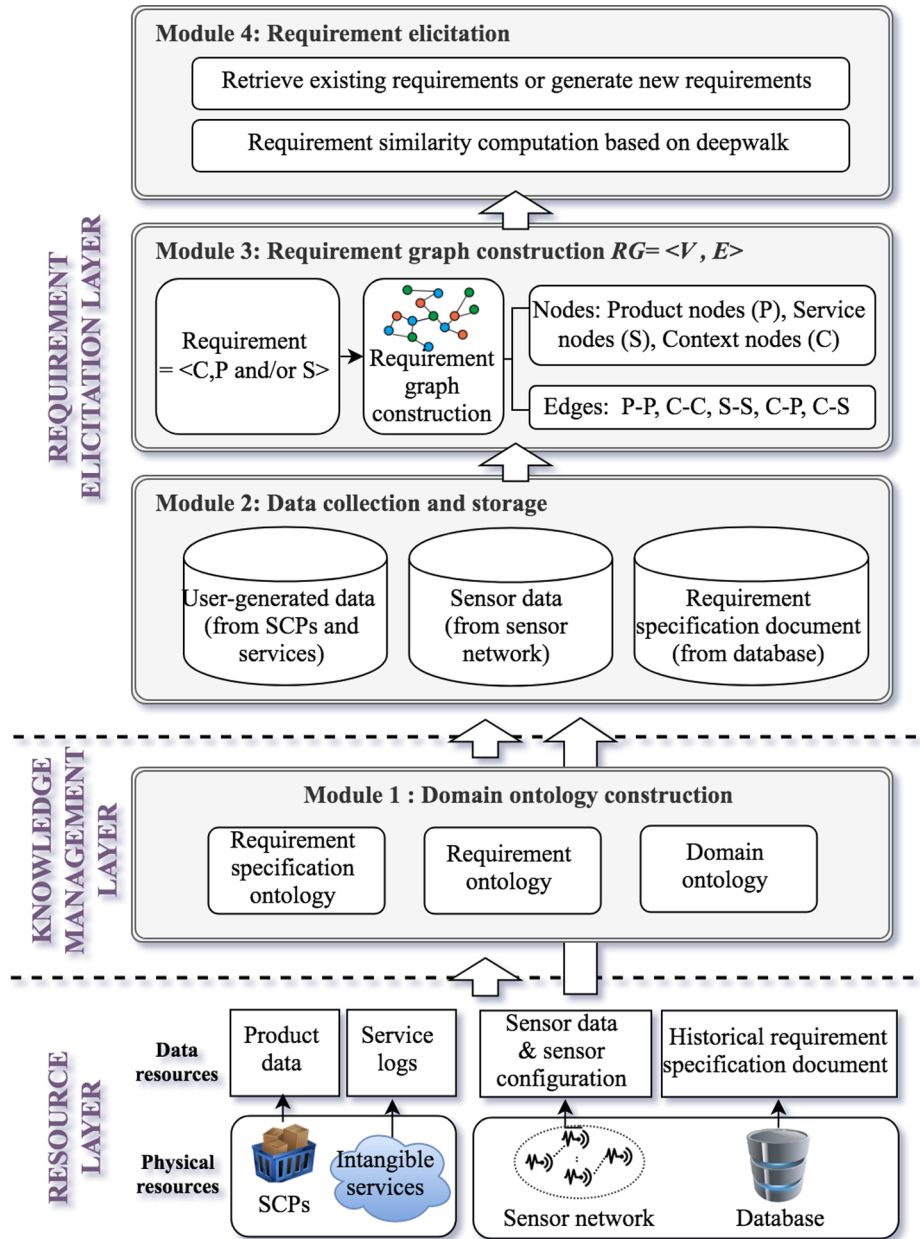


Fig. 2. The proposed requirement elicitation framework in smart PSS.

the aim of satisfying the largest populations. Li et al. [29] introduced an approach which extract project-specific and non-specific keywords from informal user request and applied machine learning algorithms to classify them. Chen et al [30] and Wang et al. [31] tried to apply graph decomposition for requirement elicitation. Mokammel et al. [32] addressed to use natural language processing (NLP) methods to automatically extract requirements lists from documents.

Moreover, requirements are usually organized in several different levels according to the degree of abstraction [33], ranging from business level and system level to feature level, function level and component level [34]. Functional requirements are the one mostly utilized and discussed in literatures. Chong and Chen [35] considered the functional requirements in the degrees of abstract level, dependency of requirements and the alternative values of requirement options. Non-functional requirements (QoS requirements) which control the performance of products or service are also frequently considered to be an equivalent optimization objective while analyzing requirements. Tripathy and Tripathy [36] developed a dynamic and fuzzy QoS

requirements-aware approach to discover services for service-oriented applications. Some researchers also expand the scope of requirements from functional requirements and/or non-functional requirements with other factors. For instance, Dey and Lee [37] considered the requirements in problem space from the perspective of user activities, environmental context and economic context.

2.3. Challenges of requirement elicitation in Smart PSS

Although the existing requirement elicitation techniques have proven their effectiveness in the context where the stakeholders can be easily identified for requirement elicitation tasks (e.g. interview, focus group, use test and so on), they are no longer capable of determining the end-user's requirements facing to the innovations of Smart PSS in a distributed IoT-enabled environment (e.g. 3D Hubs provides a cloud-based on-demand 3d printing service with a shared pool of manufacturing resources [38]). This leads to the loss of local market and the lack of deep understanding of the end-users for requirement engineers/

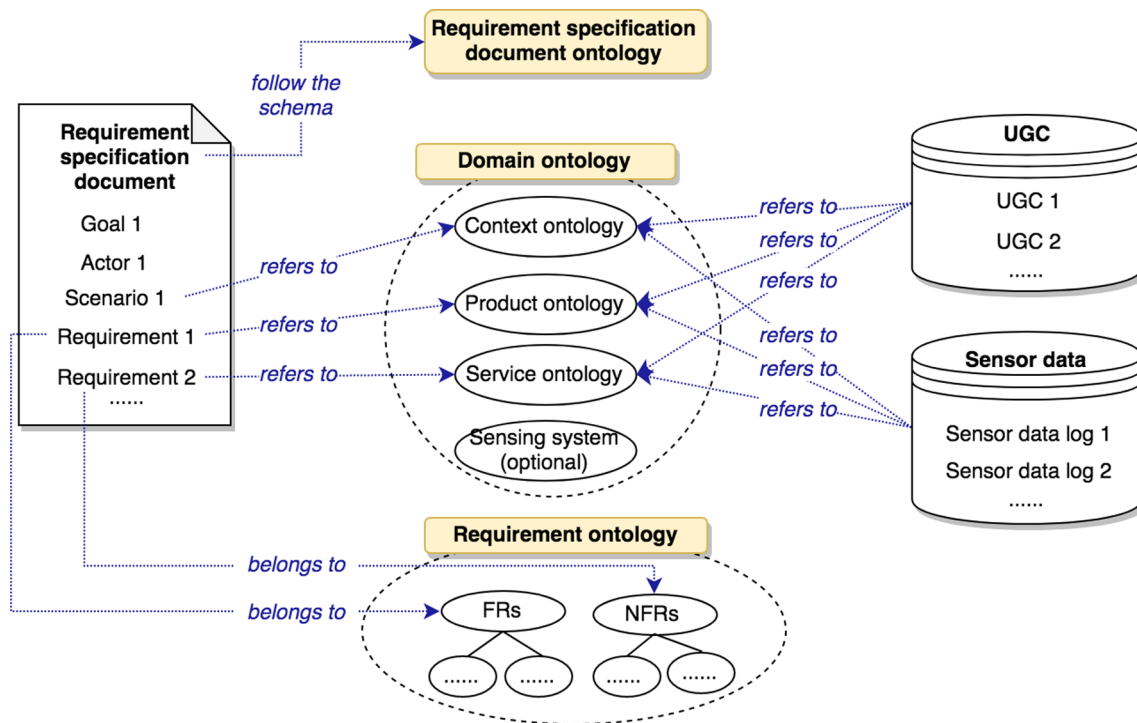


Fig. 3. The relationship of requirement specification document and ontologies.

designers [11]. Moreover, with the large-scale, heterogeneous product-sensed and user-generated data collected, the existing requirement elicitation techniques cannot manage complete the task well. A systematic approach to assist requirement engineers to analyze and manage the requirements based on the diverse data, including structural data, semi-structural and even unstructured data. Furthermore, compared with the conventional PSS which treats the development process as completed after product launched into markets rather than iterating the current PSS offerings, Smart PSS is an ever-evolving system through the upgrade of smart products or especially e-services [20,39]. In the evolving Smart PSS, the requirements are frequently changed because of technical progress or marketing. The frequently-changed requirements exist in the requirement elicitation all the time but this tendency becomes even more obvious and evident. In summary, two key challenges for modeling requirements in the Smart PSS context can be derived: (1) Massive data (e.g. product components and attribute values) are obtained in SCPs, where requirement elicitation is conducted in a big data-driven manner in the smart, connected environment, including entity content [40], textual comments [41] and environmental context [42,43]. (2) Requirement elicitation should extend to the usage stage other than just in the early design stage with context-awareness. In the Smart PSS context, user-generated data and product-sensed data are largely collected during usage for product-service evolution where context information [44,45] (e.g. location, state of people, etc.) which serve as the key to enable real understanding of user behavior and hence trigger reliable design innovation.

3. The proposed requirement elicitation framework

To overcome the abovementioned challenges, this section proposes a novel data-driven graph-based requirement elicitation framework in the Smart PSS context, as shown in Fig. 2. From the perspective of system architecture, the framework mainly contains three layers, i.e. *resource layer*, *knowledge management layer* and *requirement elicitation layer*.

In *resources layer*, SCPs and its intangible e-services, sensor networks and the database constitute the physical resources as the various

physical resources. Meanwhile, corresponding to the physical resources, diverse raw data including product data, service logs, sensor data and historical requirement specification documents can be acquired for the proposed requirement elicitation framework. The resources layer serves as the foundation of the proposed framework by supporting the following knowledge management layer and requirement elicitation layer with the heterogeneous data.

Knowledge management layer aims to provide schemas of how the knowledge is organized in the framework. Plenty of professional concepts, such as the components in SCPs and the relations between function requirements and the non-functional requirements are organized in well-defined ontologies.

Requirement elicitation layer depicts the process of integrating heterogeneous data, constructing requirement graph and finally intelligent decision-making process, where potential customer needs can be derived based on the existing product, service and context information.

To establish this proposed three-layered framework, four core modules including *domain ontology construction module*, *heterogeneous usage data collection module*, *requirement graph construction module*, and *requirement elicitation module* are further elaborated below.

3.1. Module 1: domain ontology construction

The main object of module 1 is to clarify the top-level schemas in requirement elicitation for machine-readability and define the domain concepts in domain ontology as reference for the later modules. Requirement engineering in smart PSS is a knowledge-intensive process, which highly relies on the domain knowledge of the specific project [46,47]. Ontology is used as knowledge bases in this module, owing to its good performance of representing and storing the structure of knowledge, from upper-level knowledge to the domain knowledge [26]. Based on the study of Verónica Castañeda [48], three kinds of ontologies are involved in the requirement engineering fields, i.e. requirement specification document ontology, requirements ontology and application domain ontology. Their roles and relationships are also shown in Fig. 3. To be more specific, requirement specification document ontology defines the structure of the requirement specification

document. Requirement ontology serves as the taxonomy of requirements (e.g. functional requirements, non-functional requirements and so on). Meanwhile, domain ontology serves as the structure of domain knowledge (e.g. domain knowledge concepts and their relations).

As for the establishment of these ontologies, methods with higher automation are recommended [49], and a semi-automatic method is adopted in this framework. The raw domain knowledge is extracted from websites, which contains requirement specification documents and product configuration systems with well-defined structures. Based on web mining and text mining techniques, the concepts in domains and the upper-lower relationships including Hypernym (is a kind of), Hyponym (...is a kind of), Holonym (is a part of), Meronym (...is a part of) can all be identified and organized as ontologies [50]. The concepts involved in product development engineering usually contain material, product, feature, spatial relationship, manufacturing and degree of freedom [51]. Besides, it is worth noting that the associated services [52] or even the sensing system [6] should be added into the domain ontology as well in Smart PSS because of their strong correlation with the product concepts.

3.2. Module 2: data collection and storage

3.2.1. Data collection

The heterogeneous data comes from three types of resources, namely *user-generated content (UGC)* and *sensor data and requirement specification document*.

- *UGC* denotes the user-contributed data from mobile social networking (MSN) services (online) including numbers (e.g. user ratings), structured text (e.g. tables), unstructured text (e.g. user comments and question-answering content) and images. UGC can present the human intelligence as social sensors if the meaningful concepts with semantics can be extracted from it, making it a powerful data resource for requirement elicitation. Lots of product and service features, such as high quality, fast speed, good performance can be extracted from UGC, which is useful for the context construction and requirement elicitation in the later modules.
- *Sensor data* refers to signals collected from SCPs by the embedded sensor modules. For example, acceleration, angular velocity, magnetic field, orientation, position and luminous intensity are several common parameters can be obtained by the smart mobile devices. Sensor data is used for the context construction in this framework.
- *Requirement specification document* stands for the historical requirement lists from the service provider companies. It is collected as a requirement reference for the later modules.

3.2.2. Data storage

Data will be stored in two kinds of knowledge base, i.e. relational database and NoSQL database based on its types. The integrated data can be stored in a relational database (e.g. MySQL) owing to its structural format defined above. Meanwhile, the extracted knowledge (e.g. requirement information) are stored in NoSQL database in a graph-based format, where the nodes depicts the entities with the edges to depict

their relationships. Graph is used since its essential advantage which can present the structure and proximity of nodes. It is intuitional for the subsequent knowledge discovery, for example, mining the proximity of physical components and data logs. NoSQL database is also utilized for the sake of restoring nodes and for the later reasoning applications. Neo4j, as a typical kind of NoSQL database, has the advantage of high performance for retrieval and semantization. It uses graph structure for semantic queries with nodes to represent entities and edges to represent relations. Hence, it can be well-adopted for the cases in the Smart PSS context with heterogeneous data sets.

3.3. Module 3: requirement graph construction

3.3.1. Requirement representation

While handling requirements, a requirement representation way is proposed based on a well-known requirement boilerplate, i.e. Rupp's boilerplate [53]. Requirement boilerplates, also called requirement templates or patterns, offer effectiveness in requirement engineering since they reduce the ambiguity in natural language requirements and make them more amenable for automatic analysis [54]. According to the Rupp's boilerplate, a requirement can be represented as "*Under what condition, a system shall/should/will do what process*". The '*system*' can be a product and/or a service in the context of Smart PSS and the process is usually predefined by experts or engineers. One thing worth noting that the predefined processes are constraint to the actions or process in problem domain, meaning that only user-required functionalities, such as 'be replaced', 'be maintained' and 'attach and detach' generally should be considered and predefined. In this way, the necessary and changeable parts in requirements are the '*condition*' and '*product/service*'. In this way, requirement elicitation is transformed as the task of exploring the co-occurrence relations between product information, service information and context information. In order to achieve this goal, a graph-based approach is exploited by constructing a requirement graph (RG) with three kinds of nodes, i.e. *product nodes*, *service nodes* and *context nodes* (see Fig. 4).

3.3.2. Requirement graph construction

A lexical requirement graph $RG = \langle V, E \rangle$ can be constructed where V stands for vertices of graph and E means the edges between vertices. To be more specific, V are the distinct entities in UGC and requirement specification documents, they are extracted from the raw documents based on the predefined concepts in ontologies in Module 1. E are the edges between nodes if two or more entities appear simultaneously in as sentence. Totally three kinds of nodes including product nodes (P), service nodes (S) and context nodes (C) are extracted and constructed in the requirement graph.

Product nodes (P) are entities and related auxiliary components in Smart PSS. Each entity or auxiliary component is a node. Besides items themselves as products, physical properties, such as length, width, height and weight, are typical attributes attached to the product nodes.

Service nodes (S) mostly refer to the associated services offered by service providers, such as maintenance (predictive maintenance/maintenance after broken), shipping overseas, return or revocation, as

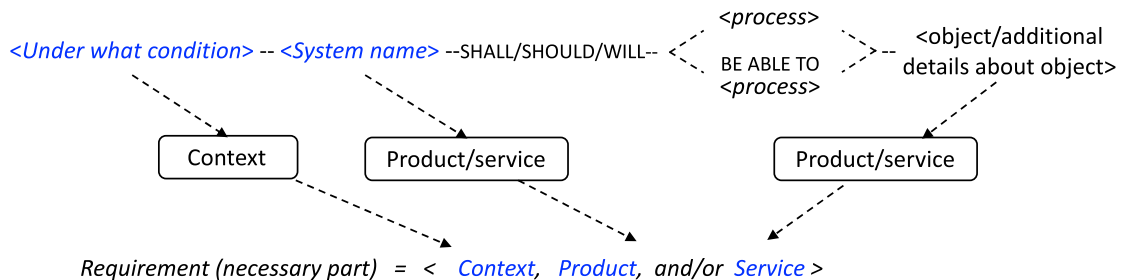


Fig. 4. The proposed requirement representation way.

Table 1
An example of the values of context factors for cycling.

Context factors	The value of context factors
Rainy or not	0: not rainy; 1: rainy
Wind speed	0: low wind speed; 1: high wind speed
Riding time	0: less than 30 mins; 1: longer than 30 mins

well as online shopping and configuration. They are configured and combined with the products and then are triggered under certain context. Product entities and service entities can be easily extracted since they are usually presented as well-structured tables with detail information by service providers.

Context nodes (C) refers to the different usage conditions which can be easily accessed through hardware sensors (product-sensed data) [55] while using the Smart PSS provisions. They are composed of factors which will affect performance of products and/or services and presented as the values of those factors. For example, the comfort of a cycling can be affected by rainy or not, wind speed and riding time, if the experts define the values of those factors as Table 1, then a “rainy” context can be presented as [1, 0, 0]. The context information can be collected from either sensor data or UGC. If it is collected from UGC, the keywords extraction techniques should be applied to extract context information. WordNet, as a commonly used and widely accepted tool to identify synsets, can be used to detect predefined keywords.

Considering all the possibility of the relationships between nodes, six types of pairwise relationships will appear in the requirement graph RG, as shown in Fig. 5. An edge can be identified if two nodes appear in a sentence simultaneously. The extracted edges are unified and undirected edges. Hence, in the proposed Context-Product-Service requirement graph, requirements can be modeled as tuples with $\langle P, C, S \rangle$ which follow the Rupp’s requirement template that “Under a certain context (condition), what products and/or service shall/should do actions” [53,22]. Each node represents a product/service/context and six types of relationship in-between them are represented by edges.

3.4. Module 4: requirement elicitation

After constructing a complex RG based on the nodes and edges, implicit requirements are supposed to be detected and elicited from the complex RG. Based on the statement of requirement in Module 3, the nature of requirement elicitation is to elicit existing co-occurrence relations and explore potential co-occurrence relations between the context nodes (C), product nodes (P) and service nodes (S). The process of requirement elicitation can be seen in Fig. 6.

Once users are using the SCPs, the condition sets can be collected periodically from sensors, together with the static product components set and the potential service set. Based on the given condition set which corresponds to a context node in RG, if it has linkage (i.e. the edge which means the co-occurrence relations) with other nodes in product components set and service set (i.e. there is an existing $\langle C, P, \text{and/or } S \rangle$ tuple), then it will be retrieved and stored in a dataset. However, another situation that the given context node does not have direct linkage with the product component set and service set might also happen.

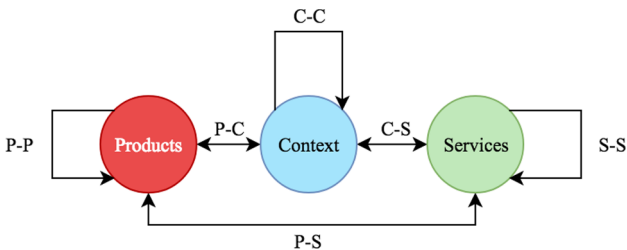


Fig. 5. Schema of Product-Context-Service requirement graph.

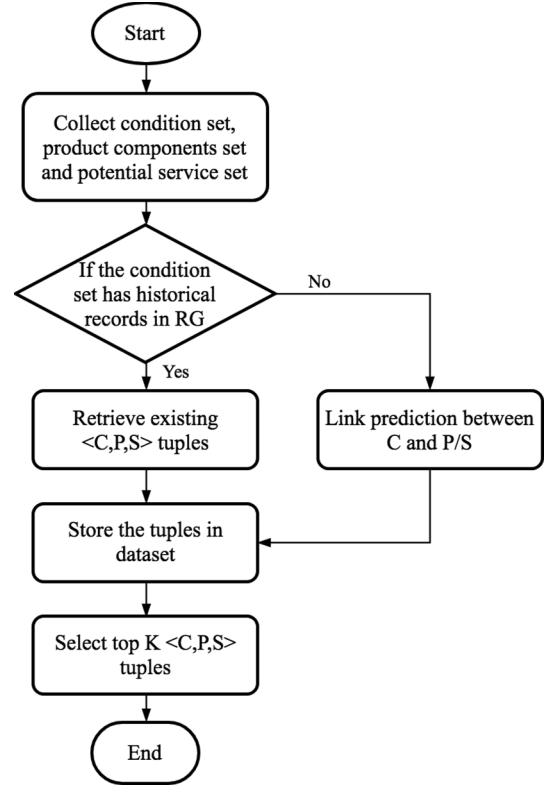


Fig. 6. Requirement elicitation process.

Though huge number of nodes can be stored in the RG, the co-occurrence information representing edges may be missed or lost during the information extraction process, or even rarely appeared due to the specialized domains, which makes the RG lack edges. This problem can be further specified as follow. Given a $RG = (V, E)$ which contains N nodes and M edges, totally $N(N - 1)/2$ potential node pairs. A score $S_{x,y}$ is given to each node pair which no linkage exists. Hence, the goal is to rank the $S_{x,y}$ based on the given context node and a set of product nodes and service nodes.

Hence, the requirement elicitation process follows a linkage prediction model, which is based on random walk method. It mainly contains two steps: (1) deriving the node embeddings by using Deepwalk model [56], and (2) computing the node similarity by utilizing cosine similarity for link prediction. The higher node similarity between two nodes refers to more possibility of a linkage between them. It is worth noting that the node similarity represents the proximity between them. The well-known cosine similarity, is the dot product of the two vectors divided by the product of the two vectors' lengths as below.

$$sim(x, y) = \frac{v_x \cdot v_y}{\|v_x\| \times \|v_y\|} \quad (1)$$

Based on the study of Perozzi et al. [56], random walk has been used to generate the sequences of nodes in a network, then these sequences of nodes will become the input of SkipGram model which can generate the node embeddings. SkipGram is originally a language model to estimate the likelihood of a specific sequence of words appearing in a corpus based on a given word. Here a node is treated as a word and the path between context nodes (C), product nodes (P) and/or service nodes (S) as the sequence of the words. The target function of its optimization model is denoted as:

$$minimize(-\log P(w_1, w_2, \dots, w_3 | w_i)) \quad (2)$$

As a result, the node similarity can be ranked based on the cosine similarity according to the node embeddings generated from Deepwalk.

Top K <C, P and/or S> tuples, which has highest probability to appear, and can be selected and recommended to requirement engineers as references for future Smart PSS provision upgrade.

4. An illustrative example

To address the above proposed framework and its requirement elicitation process, an illustrative example of smart bicycle reverse design was discussed. Since cycling is becoming an increasingly popular sport and can show the healthy lifestyle of a person, more and more people prefer personalized bike which can almost perfectly fit their body and their lifestyle [57]. This example was conducted based on the open data collected from a bicycle component website of a Germany company [58], which sells bicycles and its components together with e-services provided, such as maintenance, revocation, return and etc. in a smart connected environment.

4.1. Domain ontology construction

In terms of the statement in Section 3, ontologies, especially domain ontologies, are established to manage the domain knowledge and serve as the schema of concepts in domain. Web mining and data mining techniques are utilized while extracting meaningful concepts and building ontologies. Since the website contains rich information of bicycle components and services, the product ontology and the service ontology can be established based on the structure of the website navigation, as shown in Fig. 7(a) and (b), which is constructed via Protégé. Meanwhile, the contexts (i.e. use scenarios in this example) of riding bicycle are also discussed by two bicycle designers. The contexts are subdivided based on the cycling operational sequence and represented as a hierarchy of sub-scenarios in Fig. 8. According to the use scenarios, the context ontology of riding bicycle is built and shown as Fig. 7(c).

4.2. Data collection and pre-processing

In this example, heterogeneous data are collected from UGC, sensor data and historical requirement specification documents which are depicted as follows.

4.2.1. UGC

The user-generated contents are defined as the user ratings and the comments to the bicycles and the related components from the website. They are represented as natural language and numbers. The UGC are sparse that most of bicycles and components do not have any comments, this phenomenon will be even more significant if the domain is more specialized. A web crawler software assisted the data collection process based on the given URLs. All the user comments were stored in a csv file. Totally 4392 pieces of user comments are collected.

If more than one concepts or entities (i.e. product components, service and context) appear together in a user comment, then there is a relation between the entities. Hence a text mining process is deployed to extract the entities from UGC. The specific keywords of concepts of entities have already been identified in the domain ontologies.

4.2.2. Sensor data

Sensor data is the main data resource to constitute different context. Naturally, the sensor data contain heterogeneous data. Because of the lack of sensor data, only three kinds of sensor data are chosen as context factors, shown as Table 2. The factor of “Riding distance” and “Rainy” can be detected through the bicycle computers which can record cycling distance and the weather. The saddle pressure can be collected via the seat sensors embedded in the saddles. Their modeling functions are defined by experts.

4.2.3. Requirements from requirement specification document

Based on the use scenario and the discussion between experts, some requirements of riding bicycle can be identified as the minimum information for the early design. They are stored in the requirement specification document and service as one types of information source

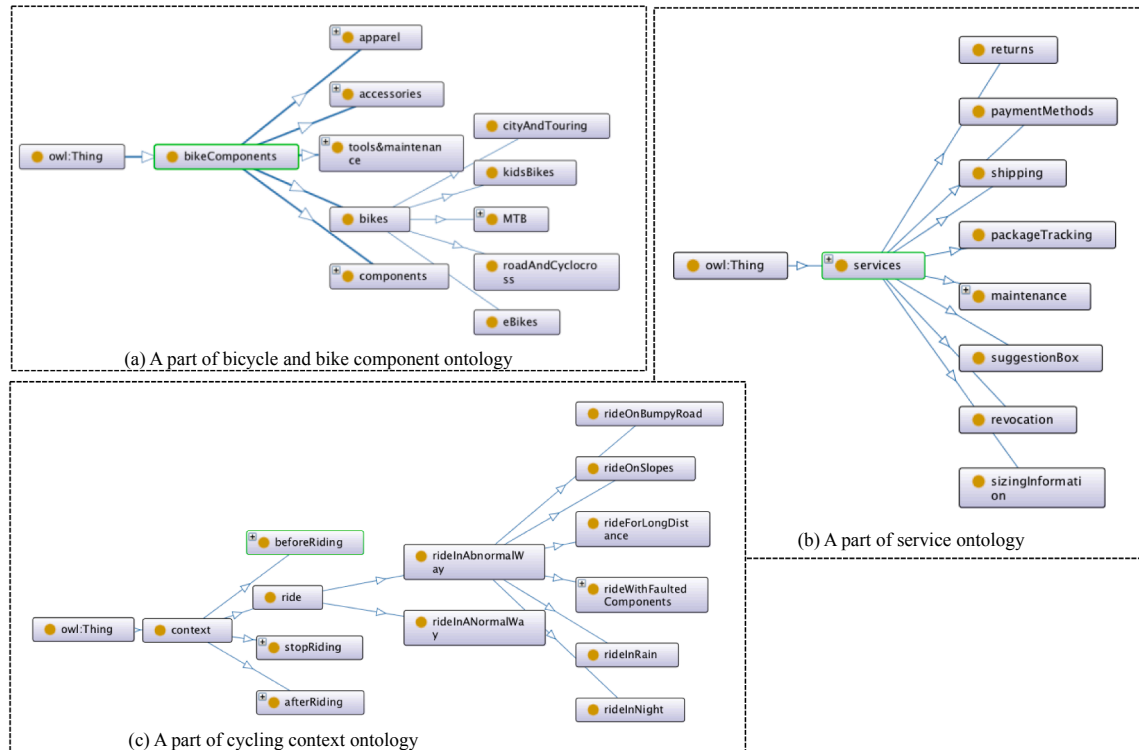


Fig. 7. Domain ontologies of bicycle.

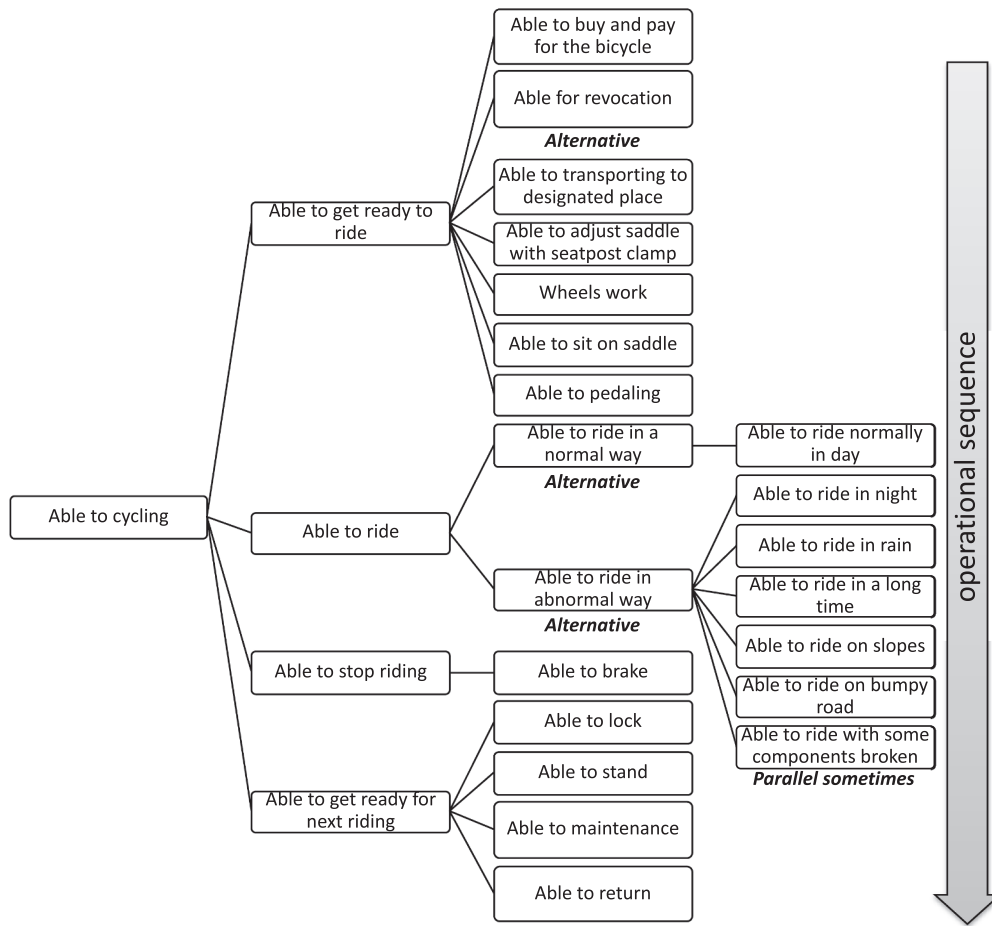


Fig. 8. An example of use scenario of riding bicycle.

Table 2
Context factors for cycling.

No.	Context factor	Modeling function
1	Riding distance	0: less than 300 km; 1: more than 300 km
2	Saddle pressure	0: less than 200; 1: more than 200
3	Rainy	0: not rainy; 1: rainy

Table 3
A part of initial requirements.

Requirement list
1. A bicycle splashguard should be easy to attach
2. A bicycle splashguard should be easy to detach
3. A bicycle chain should be fast to attach
4. A bicycle chain should be fast to detach
5. The interface of splashguard with bike should not mar
6. The interface of splashguard with bike should not catch water
7. A bicycle saddle should be intact
8. A bicycle saddle should be attached to the seatpost
9. A bicycle saddle can be adjusted by the seatpost clamp
10. A bicycle frame should not bend
11. A bicycle wheel should fit into the frame and fork
12. A bicycle wheel can hold bicycle tyre
13. A bicycle tyre should be easy to replace
14. A bicycle tyre should be lightweight
15. A bicycle handlebar should provide a mounting place for brake levers, shift levers, bicycle computers and bells
16. A bicycle brake can reduce the speed

in this example. Table 3 shows 16 pieces of requirements in requirement specification document, which follow the Rupp's requirement boilerplate.

4.3. Requirement graph construction

Three types of information are treated as three sets of nodes, *P*, *S* and *C* to build a requirement graph.

Product nodes (P) are product components collected from the bicycle website, including brake, chain, crankset, saddle and so on. Each product component is treated as a product component node with some attributes, for instance their full name and price.

Service nodes (S) refers to existing services which can be provided to users. The service components are provided by the Germany company and they are described by text in this example. Five types of services are involved, namely 'shipping', 'maintenance', 'returns', 'payment methods' and 'package tracking'.

Context nodes (C) means the context with evolving attributes of bikes during usage phase, such as saddle pressure and total cycling distance. They are made up of various sensor data. The contexts have been already predefined in the context ontology. Among all the contexts, only three types of context are selected for further analysis because they are typical scenarios in cycling and can be linked with the sensors embedded into smart bike.

All the involved entities for the requirement elicitation are analyzed first. In this example, 597 bicycle components and 5 types of services are collected together from the website of Germany bicycle company [58]. Three kinds of context are assumed by two domain experts who had bike design experience to run the example due to the lack of

Table 4
Statistical information of the dataset.

Nodes/relationship	Entity/relationship	Sample size	Average degree
C	Context	3	65.3333
P	Product	597	2.2288
S	Service	5	277.4561
C-C	Context-Context	1	
C-P	Context-Product	392	
C-S	Context-Service	8	
P-P	Product-Product	/	
P-S	Product-Service	1054	
S-S	Service-Service	4	

practical context information. Both the context nodes and service nodes have high average degree, indicating high co-relationship among them, while product nodes are not connected densely. All of the data are stored in the graph database, i.e. Neo4j and the statistical information of the dataset can be found in Table 4.

4.4. Requirement elicitation

The goal of this phase is to project the nodes into a 64-dimensional space and then extract the similarity between nodes. By leveraging Deepwalk, the 605 nodes with edges are embedded into a d -dimensional space $\mathbb{R}^d$. The learned representation in $\mathbb{R}^d$ can be easily conducted by computational operation. As already mentioned, cosine similarity between two nodes are computed to represent the correlation of nodes. The codes are completed by Python on Jupyter Notebook. Given the id of input node, the approach is able to return the other two connected nodes as tuples. Tuples are ordered by similarity high to low. The result of the ranked similarity between nodes are shown in Table 5. It is clearly to see that all the cycling contexts are highly related with product concepts. No services appear in the results, meaning that the services provided by the company do not significantly influence the context during usage stage. Thus, more services which can be combined into the usage stage should be considered by the company. Furthermore, the concepts are embedded into the Rupp's template as the final results and assist requirement engineers for further developments. Table 6 represents the extracted requirements, in which the blue parts are the extracted concepts. It is worth mentioning that this example is designed to find out which product components and/or services have opportunities to be improved instead of generating the design solutions directly. Hence all the actions in requirement are predefined as "be upgraded".

Table 5
Top 5 tuples for each context node.

Context nodes	Connected nodes	The concept of the connected nodes	Similarity
Ride for long distance	Shimano CN-HG93 9-speed Chain	Chains	0.9591
	Campagnolo Neutron Ultra Wheelset	700c (28") Wheels	0.9504
	SRAM GX Eagle 12-speed XG-1275 Cassette + PC GX Eagle Chain Set	Wear & Tear Sets	0.9453
	Continental Speed Ride ProTction 28" Folding Tyre	700c (28") Tyres	0.8273
	Brooks B17 Standard Saddle	Saddles - Comfort	0.8095
Saddle is not comfortable	SDG Ti-Fly Storm Sam Hill Saddle	Saddles - MTB	0.9215
	Brooks Cambium C17 Carved All Weather Saddle	Saddles - Comfort	0.9170
	Brooks Swift Titanium Saddle	Saddles - Comfort	0.8244
	Brooks B17 Imperial Saddle	Saddles - Comfort	0.7648
	SQlab 611 Ergowave active Saddle	Saddles - MTB	0.7452
Rainy	Shimano Stainless Steel Shifter Cables - 100 Pack	Cables	0.9086
	Maxxis Highroller II 3C MaxxTerra 27.5" Folding Tyre	27.5" Tyres	0.8934
	Fulcrum Racing Sport Wheelset	700c (28") Wheels	0.8896
	Tektro Disc Brake Pads for Auriga/Orion/Draco/TRP HY/RD	Disc Brake Pads	0.8880
	Magura Type 9.C Comfort Brake Pads for 4 Piston MT Disc Brakes	Disc Brake Pads	0.8858

Table 6
The results for extracted requirements.

No.	Requirements in requirement template
1	When "ride for long distance", chains should be upgraded
2	When "ride for long distance", 700c (28") Wheels should be upgraded
3	When "ride for long distance", Wear & Tear Sets should be upgraded
4	When "ride for long distance", 700c (28") Tyres should be upgraded
5	When "ride for long distance", Saddles - Comfort should be upgraded
6	When "saddle is not comfortable", Saddles - MTB should be upgraded
7	When "saddle is not comfortable", Saddles - Comfort should be upgraded
8	When "rainy", Cables should be upgraded
9	When "rainy", 27.5" Tyres should be upgraded
10	When "rainy", 700c (28") Wheels should be upgraded
11	When "rainy", Disc Brake Pads should be upgraded

5. Discussion

In Section 2, two major challenges of requirement elicitation tasks in Smart PSS are defined as: (1) *big data-driven in a smart, connected environment*, and (2) *context-awareness during usage stage*. Regarding to the prior one, this example collected large-scale heterogenous data sources including both UGC (e.g. textual data, semi-structural table data) and product-sensed data, which is more compatible and reliable compared to the existing works only considering the textual information [59–61]. Meanwhile, the requirement elicitation process considers the context information by establishing the <C, P and/or S> model, which can deal with the change of requirement with context-awareness. If a new or similar context happen in the requirement graph, the framework can still provide predictions as the potential requirements for requirement engineers to conduct further actions.

Despite its advantages, the above example has some limitations. Restricted by the essence of existing open datasets, the context nodes are manually generated by two experts who have experience of bike design instead of collecting practical data from real industrial cases. Besides, the algorithm for link prediction in Module 4 is only based on the paths in the RG. In fact, other kinds of data, for instance the semantic information of product nodes based on product concepts can also assist to find the proximity between nodes. This kind of information should be considered and added in the future works about the efficiency of requirement elicitation. Nevertheless, in general, these limitations do not affect the validation process of the proposed requirement elicitation framework in Smart PSS.

6. Conclusion

Recent rapid development of ICT and AI techniques have enabled a novel value co-creation proposition, i.e. Smart PSS. It utilizes SCPs as the medium and tool, the intelligent system as the infrastructure,

stakeholder as the co-creation participants, and their generated e-services as the key values delivered that continuously strives to meet individual customer needs in a sustainable manner. In this complex system, massive user-generated/sensed data plays a critical role to enable the final design innovation success. Hence, how to effectively and accurately elicit stakeholders' requirements is of paramount importance. This work, for the first time, introduces a novel data-driven graph-based framework to support requirement elicitation process for design innovation in the Smart PSS context. The main scientific contribution of this research can be summarized into three aspects:

- (1) Well-defined the uniqueness of requirements elicitation in the Smart PSS context. In Smart PSS, the process of requirement elicitation is different from conventional RE process, mainly in two aspects: a) *big data-driven in a smart, connected environment*, and b) *context-awareness during usage stage*.
- (2) Presented a generic and compatible framework to handling requirements in the Smart PSS context. $\langle C, P \text{ and/or } S \rangle$ tuples treat the key information in requirements as nodes and allow the attributes and other associate information attached to the nodes, reducing the constraint that only one type of data can be used for requirement elicitation process.
- (3) Proposed a systematic data-driven graph-based approach for requirement elicitation process. Instead of conventional requirement elicitation framework dealing with explicit stakeholders' expression, requirements are solicited in a data-driven manner objectively by leveraging the context information. Furthermore, the requirement elicitation task is defined as a linkage prediction problem, which provides a new perspective to deal with the requirement elicitation task.

Moreover, an illustrative example of smart bicycles design improvement is further adopted to validate the feasibility of the proposed data-driven requirement elicitation framework. The authors hope this framework can be seen as the premise and foundation to support the requirement elicitation, and further value co-creation process in the Smart PSS context. Meanwhile, it is recommended that more in-depth research should be conducted in the future to: (1) consider semantics similarity while computing the nodes similarity, (2) develop a robust system platform for various cases implementation, and (3) further study the advanced algorithms for better robustness of requirement elicitation process.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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